

equicenta® CTM vs. Pro-Stride® APS

Two very different products often considered for the same joint cases. This sheet lays out how they compare on mechanism, indications, evidence base, and the practical decisions vets and clients care about.

KEY REFERENCE

Bertone AL, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses. *Frontiers in Veterinary Science* (2026). DOI: [10.3389/fvets.2026.1833933](https://doi.org/10.3389/fvets.2026.1833933)

	equicenta® CTM Cryopreserved equine umbilical cord tissue matrix (biologic allograft)	Pro-Stride® APS Autologous protein solution (blood-derived orthobiologic device, Zoetis)
Category	equicenta® CTM	Pro-Stride® APS
Product class What it is	Allogeneic biologic. Cryopreserved equine umbilical cord tissue matrix containing native extracellular matrix proteins, growth factors, and extracellular vesicles. Off-the-shelf.	Autologous blood-derived device. Two-step processed concentrate of the horse's own plasma, platelets, and white blood cells, enriched in anti-inflammatory cytokines and growth factors. (Zoetis product literature)
Mechanism of action Primary MOA	Biologic: supports tissue repair and modulates inflammation via a multi-modal payload of native ECM proteins, growth factors, and small "messenger" particles released as the matrix biodegrades.	Cytokine-blockade + growth factors: concentrates IL-1RA (IRAP), A2M, soluble IL-1R and TNF-R, plus platelet-derived growth factors, to inhibit IL-1/TNF-α inflammatory pathways in the joint. (Zoetis synovitis brochure)
Site of action Where it works	Acts across joint tissues (synovium, cartilage matrix, peri-articular soft tissue) via biologic signaling; also used peri-tendinously and peri-ligamentously.	Intra-articular use only. Additional in-clinic use cases described include joint synovitis, tenosynovitis, and navicular syndrome, but joint injection is the studied indication. (Zoetis product page; KVA Equine practice literature)
Labeled / indicated uses Where each is used	Studied in a mixed real-world population of joint, tendon, and ligament conditions in performance horses across 8 U.S. equine specialty practices.	Marketed as a device for managing equine osteoarthritis and joint lameness. Regulated as a device, not a drug. (Zoetis product page)
Tissue scope in evidence Joints vs. soft tissue	Joints, tendons, and ligaments reported in the same pragmatic controlled trial (195 conditions).	Published equine efficacy centered on joints. Anchor RCT evaluated a single joint (middle carpal / metacarpophalangeal). Broader soft-tissue case-series exists in practice but is not the strongest evidence tier.
Autologous vs. allogeneic Consistency of starting material	Allogeneic, standardized. Same starting material every dose — removes patient-to-patient variability from older, metabolic, or heavily pre-treated horses.	Autologous. Made from the individual horse's blood at time of treatment. Composition varies with the donor horse's age, health, hydration, and comorbidities.
Peer-reviewed evidence Anchor publications	Characterization & 42-day biocompatibility (BMC Medicine 2025); EV/exosome characterization (ACVSMR 2026 poster); Bertone et al. 2026 pragmatic controlled multicenter clinical trial (Frontiers Vet Sci).	Bertone & Ishihara 2014 AJVR — randomized saline-controlled RCT, n=40, single joint, 14-day primary endpoint (client assessment at 12 & 52 wk). Usimaki/Watkins 2024 EVJ — IL-1β induced synovitis model, n=18. Human/knee OA APS literature (nSTRIDE) also cited by Zoetis.
Study design context How the evidence was generated	Pragmatic multicenter controlled trial: 138 horses / 195 conditions / 8 practices; natural-disease enrollment; open-label; active-comparator vs. clinician-selected standard of care; blinded statisticians & imaging readers.	Anchor RCT: single-site, saline-controlled, single-joint, 14-day primary endpoint (short-horizon efficacy signal); n=40. Watkins 2024: induced-disease model, n=12 treated + 6 controls. Neither study evaluated tendon/ligament outcomes or provided a 26-week active-comparator return-to-performance signal.
Clinical outcome signals How improvement is characterized	Faster and greater improvement in lameness, pain, and swelling vs. SOC; arm-level return-to-performance 52.9% vs. 29.3% at 26 wk (OR 2.72; p<0.006). Owner-reported body condition, coat, overall health, and QoL also favored eCTM (p<0.03).	Bertone & Ishihara 2014: APS improved lameness grade and kinetic gait vs. saline at 14 days; response was concentrated in horses with lameness <4 and minimal radiographic OA. Watkins 2024: APS reduced synovitis pathology scores but did not significantly reduce lameness or joint circumference vs. untreated controls in the IL-1β model.
Onset & durability Timing profile	Separation from SOC by week 2 (pain, swelling) and week 4 (lameness); sustained through 12 wk (p<0.001); performance signal durable at 26 weeks.	Anchor RCT primary endpoint at 14 days; client-assessed improvement reported at 12 and 52 weeks. Commercial materials describe effects lasting up to one year. No published 26-week active-comparator return-to-performance data in the peer-reviewed equine record.

Category	equicenta® CTM	Pro-Stride® APS
Dosing & workflow Rep-relevant practical facts	Off-the-shelf, ready-to-use, single-dose in most cases. No blood draw, no in-clinic processing time. Immune-privileged allogeneic tissue.	In-clinic autologous processing: ~20-minute two-step process on the Zoetis Universal Centrifuge from a fresh blood draw. Same-appointment harvest, processing, and injection. Practice must own the Zoetis centrifuge and the disposables kit. (Zoetis product page; KVA Equine)

Category	equicenta® CTM	Pro-Stride® APS
Safety profile Comparative safety	Peer-reviewed 42-day biocompatibility (BMC Medicine 2025). In Bertone 2026, eCTM had fewer injection-related adverse events than the clinician-selected SOC arm (standardized public figure: 4.8× fewer injection reactions).	Bertone & Ishihara 2014: no adverse effects reported with APS at 14 days. Watkins 2024: no adverse effects in treated horses. Broader clinical use has been characterized as safe. As with any intra-articular injection, iatrogenic risks (sepsis, flare) apply. (Bertone-Ishihara 2014 AJVR; Usimaki 2024 EVJ)
Regulatory status (U.S.) How each is positioned	Working with FDA CVM to open the NADA; product currently manufactured under cGTP with transition to cGMP by end of Q3.	Commercially distributed in the U.S. as an autologous device (Zoetis); point-of-care processing kit + centrifuge platform. Not FDA-approved as a drug. (Zoetis product page)
Consistency & QC What each dose looks like	Batch-tested finished product with a defined proteomic and EV profile from a closed-herd source (Coteau Grove Farms).	Final composition depends on the individual horse’s blood at time of collection. Zoetis has published data showing NSAIDs do not significantly alter the cytokine/growth-factor output; other patient variables (age, metabolic status, prior therapies) are not similarly characterized. (Zoetis synovitis brochure)
Practice economics Rep talking points	No capital equipment required. Single ready-to-use dose. Straightforward billing per treated horse.	Requires the Zoetis Universal Centrifuge (capital equipment) plus the per-treatment Pro-Stride disposables kit. Chair time for blood draw + processing before injection.

When to lead with equicenta® CTM	When Pro-Stride® APS may be the better fit
• Case involves tendon or ligament alongside joint disease — APS has its published efficacy in joints; eCTM has published outcomes across joints, tendons, and ligaments in the same trial. • Older, metabolic, or heavily pre-treated horse where autologous yield is likely to be variable — an off-the-shelf standardized allograft removes patient-side variability. • Advanced or radiographically significant OA — Bertone & Ishihara 2014 reported APS response concentrated in horses with lameness <4 and minimal radiographic OA. eCTM was studied across a broader case spectrum. • No in-clinic processing time or the practice does not want to invest in the Zoetis centrifuge platform. • Client wants a treatment that is peer-reviewed against a real-world active comparator arm, not a saline placebo, over 26 weeks.	• Clean, single-joint OA in a horse with a favorable Bertone-Ishihara 2014 responder profile (lameness <4, limited radiographic changes). • Vet has an existing Zoetis Universal Centrifuge and Pro-Stride kit and the case is well-suited to same-appointment autologous processing. • Owner or vet has an explicit preference for an autologous, blood-derived product rather than an allogeneic tissue biologic. • Case is a corticosteroid-eligible OA where the vet wants to avoid steroids for metabolic reasons and is choosing between APS and PRP. • Practice workflow supports a longer chair time for blood draw + processing before the injection.

30-second positioning: “equicenta CTM and Pro-Stride APS are often considered for the same joint cases but they’re fundamentally different tools. Pro-Stride is a same-day autologous concentrate of the horse’s own IL-1RA, A2M, and growth factors — the composition depends on that individual horse’s blood on that day. equicenta CTM is an off-the-shelf allogeneic biologic with a consistent proteomic and EV profile from a closed herd, published clinical evidence across joints, tendons, and ligaments in a 138-horse, 8-practice controlled trial, and a 26-week performance signal against real-world standard of care. If the case is a clean single-joint OA in a healthy horse and the practice already owns the Zoetis platform, Pro-Stride is a reasonable choice. If tendon or ligament is involved, if the horse is older or heavily pre-treated, or if the case has meaningful radiographic OA, lead with equicenta CTM.”

APPROVED WORDING RULES (Pro-Stride conversations)

SAY	DON'T SAY
“Pro-Stride is an autologous blood-derived device; equicenta CTM is an allogeneic tissue biologic. Different tools for different cases.”	“Pro-Stride doesn’t work” — Pro-Stride has published efficacy in its indicated joint use (Bertone & Ishihara 2014, AJVR).
“Autologous products depend on the individual horse’s blood quality; eCTM removes patient-side variability with a standardized, batch-tested tissue product.”	“Autologous products are inconsistent” as a blanket statement — say “composition varies with the donor horse’s age, health, hydration, and comorbidities.”

"The Bertone-Ishihara 2014 APS response was concentrated in horses with lameness <4 and minimal radiographic OA — worth considering when case-selecting."	"APS only works in mild cases" as a summary — quote the responder-subgroup finding directly from the paper.
"In the Watkins 2024 IL-1 β synovitis model, APS improved joint pathology scores but did not significantly reduce lameness or joint circumference vs. controls."	"APS failed in the induced-synovitis study" — describe the finding factually and attribute to Usimaki/Watkins 2024 EVJ.
"In our multicenter controlled trial, eCTM outperformed clinician-selected standard of care on lameness, pain, swelling, and return to performance."	"eCTM beat Pro-Stride head-to-head" — the Bertone 2026 trial did not run a direct head-to-head vs. Pro-Stride; APS was one of several products a vet could select inside the SOC arm.
"equicenta CTM is off-the-shelf with no capital equipment required; Pro-Stride is a same-appointment autologous workflow that runs on the Zoetis Universal Centrifuge."	Disparaging Zoetis, the Universal Centrifuge platform, or the practice's existing investment in it.

Sources. Pro-Stride product characterization, MOA, and workflow: [Zoetis Pro-Stride APS product page](#), [Zoetis Pro-Stride synovitis brochure](#). Anchor efficacy RCT: Bertone AL, Ishihara A, et al. 2014, [AJVR PMID 24471750](#). Induced-synovitis study: Usimaki A, Watkins A, et al. 2024, [Equine Vet J](#). equicenta CTM clinical evidence: Bertone AL, et al. 2026, [Frontiers in Veterinary Science](#), DOI [10.3389/fvets.2026.1833933](#). Characterization & 42-day biocompatibility: [BMC Medicine](#) 2025. EV/exosome characterization: [ACVSMR 2026 poster](#).

Disclaimer. Pro-Stride®, Restigen®, and Zoetis® are registered trademarks of Zoetis Inc. or its affiliates. Use of Pro-Stride product information here is for comparative reference under fair-use vet-education purposes and does not imply endorsement or affiliation. equicenta® CTM is a trademark of Equine Performance Labs. All comparative statements are based on publicly available product literature and peer-reviewed publications; the Bertone 2026 trial did not evaluate Pro-Stride as a named head-to-head comparator arm — APS was one of several products vets could select inside the clinician-selected standard-of-care arm. Individual results may vary; product selection is a clinical decision for the treating veterinarian.